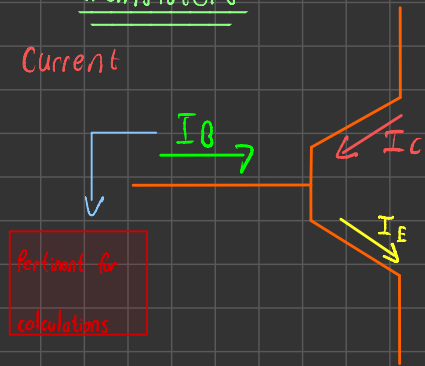


Transistors

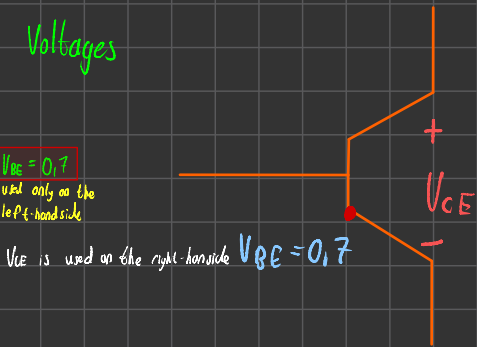
Current



$$I_C = \beta I_B$$

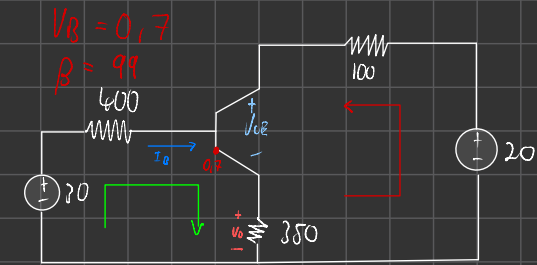
$$I_E = (1 + \beta) I_B$$

Voltages



$V_{BE} = 0.7$
used only on the left-hand side

V_{CE} is used on the right-hand side



KVL

loop 1

$$-30 + 400 I_B + 0.7 + 350 I_E = 0$$

$$-30 + 400 I_B + 0.7 + 350 (1 + \beta) I_B = 0$$

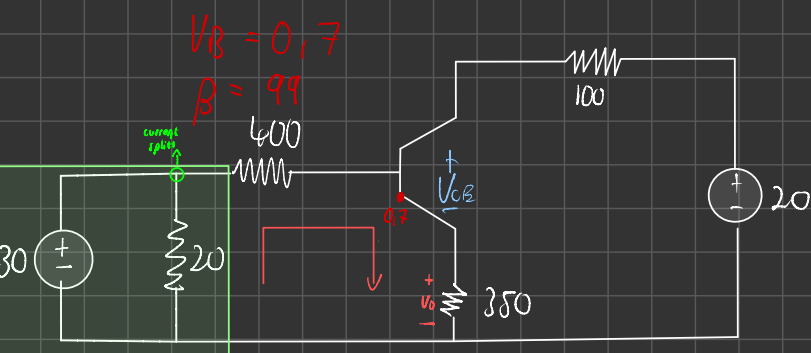
$$I_B = 8.27 \mu A$$

$\Rightarrow V_0 = ?$

$$V_0 = \pm [I R]$$

$$= + (I_E \times 350)$$

$$= (100) I_B \times 350$$

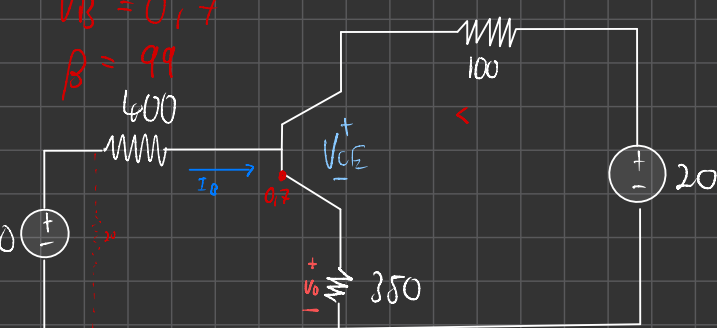


voltage is constant

$$V_0 = 70V$$

$$V_{BE} = 0.7$$

$$\beta = 99$$

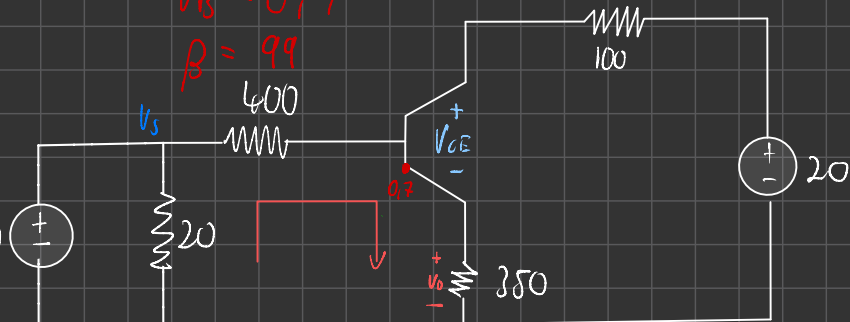


remove 20

$$V_0 = 70V$$

$$V_{BE} = 0.7$$

$$\beta = 99$$



$$V_0 = I_E R$$

$$70 = (1 + \beta) I_B \times 350$$

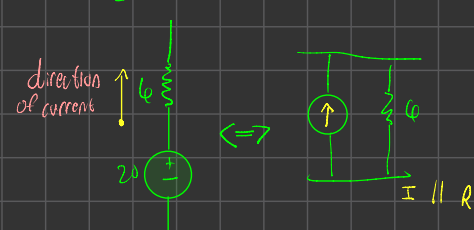
$$I_B =$$

$$-V_S + 400 I_B + 0.7 + V_0 = 0$$

Chapter 4 : Source Transformation

Series

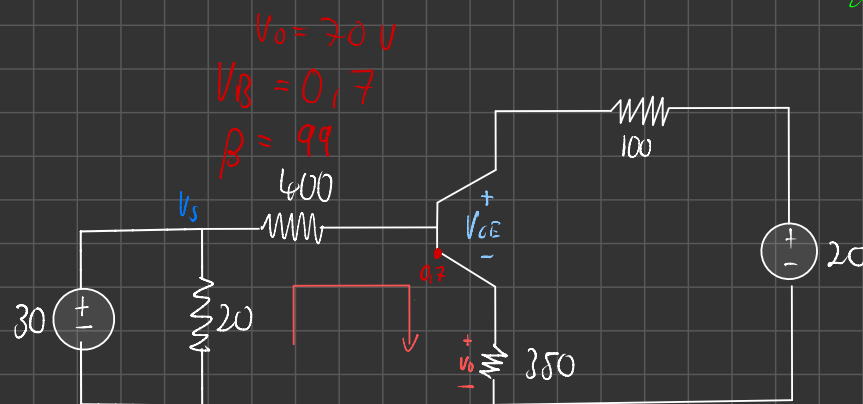
Parallel



$$V = I R$$

$$I = \frac{V}{R} = \frac{20}{4} = 5$$

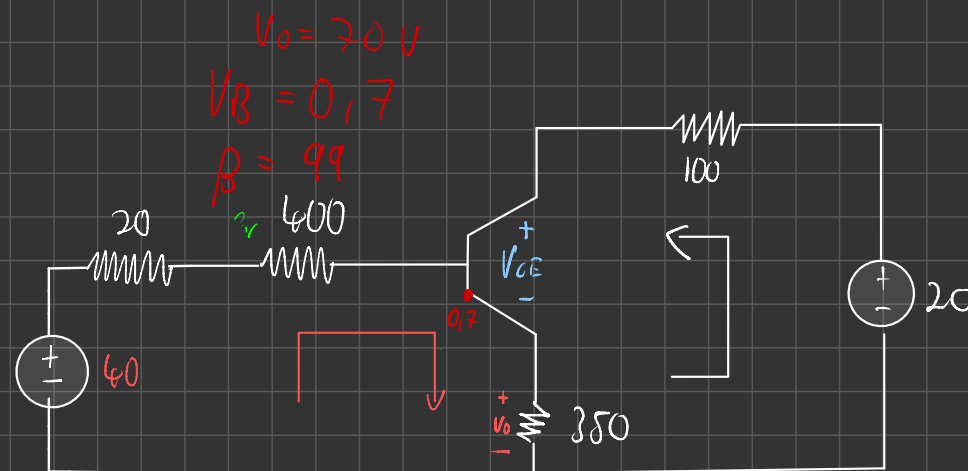
Voltage source to current source only



$$V_0 = 70V$$

$$V_{BE} = 0.7$$

$$\beta = 99$$



$$V_0 = 70V$$

$$V_{BE} = 0.7$$

$$\beta = 99$$

$$V = 2 \times 20 = 40$$

$$-40 + 20 I_B + 400 I_B + 0.7 + 350 I_E = 0$$

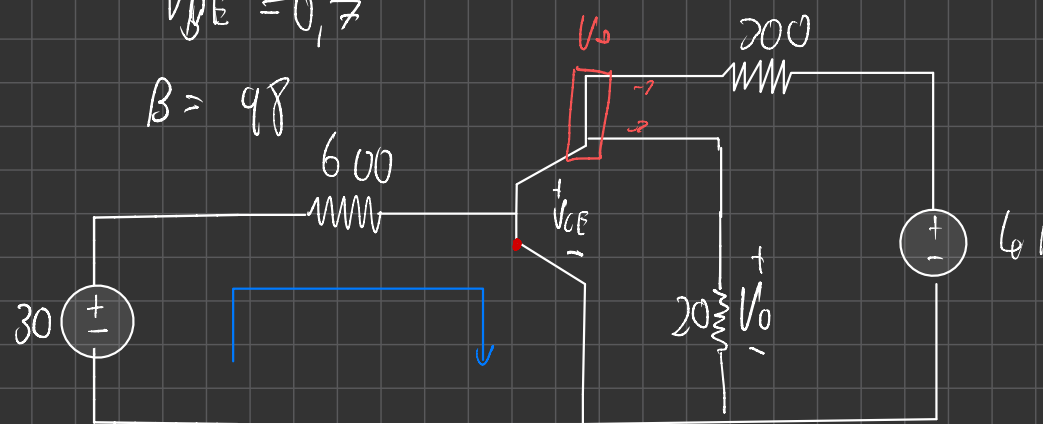
$$((1 + \beta) I_B)$$

$$I_B = ?$$

$$V_0 = (1 + \beta) I_B \times 350$$

$$V_{BE} = 0.7$$

$$\beta = 98$$



$$V_0 = ?$$

$$V_0 = ?$$

$$-30 + 600 I_B + 0.7 = 0$$

$$I_B = 48.83 \text{ mA}$$

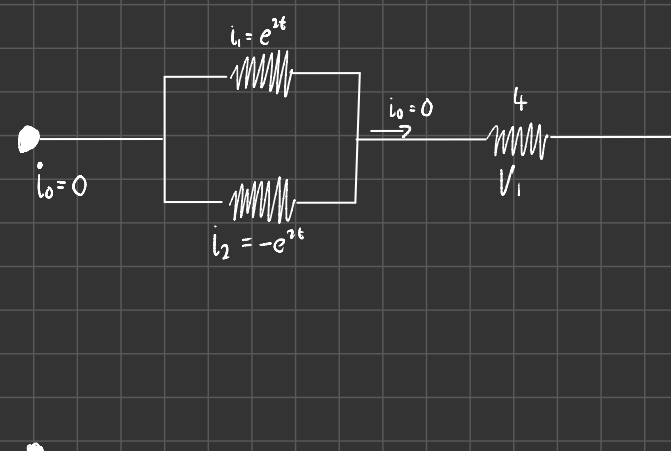
$$\frac{V_0}{20} + \frac{V_0 - 61}{300} + I_C = 0$$

$$\beta I_B = 0$$

Open Circuit

$$R = \infty$$

$$R = 0$$



$$V_1 = I R = 0 \times 4 = 0V$$